AI

Full stack

Web3

Devops

AI

1. History - what is ai, how did we end up at trnsfmrs. Fast tracking to AI
2. History - what is DL, backprop NLP,

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1. Neural networks, Pytorch
2. Optional extra class - RNNs, LSTMs, Sequential models
3. Optional extra class - CNNs
4. Coding simple attention.
5. Vanilla attention to industry -- Coding other variations of attn -
   Adding Kv cache, MQA, GQA, (Grouped q atn), MLA (multi-head latent attention), DSA
6. Huggingface end to end - datasets, spaces, models, blogs, courses etc.
7. Instrumenting LLM calls/observability/tracing
8. Vector DBs and RAG
9. Context engineering - Summarization, data collection.
10. Agents from first principles. Building an agent framework.
11. Agent frameworks
12. Memory
13. MCP
14. Computer use & multimodal agents
15. What is Finetuning
16. Finetuning a model for any usecase
17. RL fine tuning.
18. Evals -- Testing agents.
19. Advance topics
20. Other tangents --- Voice, image, video

Projects
1. Agent framework
2. RL Finetuning project + writing evals
3. Devin
4. Memory framework

Resources
1. Coursera (free)
2. https://cognition.ai/blog/1
3. https://www.anthropic.com/engineering
4. Andrej karpathy YT channel
5. 3b1b
6. https://blog.langchain.com/

Full stack

1. HTML/CSS
2. JS Basics
3. JS architecture
4. Async JS
5. Node vs Browser JS
6. HTTP and Express
7. Databases and Mongo
8. Postgres + Prisma/drizzle
9. Typescript
10. Turborepo
11. BunJS
12. React
13. Tailwind
14. NextJS
15. Websockets + WebRTC
16. Queues/Pubsubs

Projects
1. todo app
2. Lovable (easy)
3. codeforces
4. Trading app

Resouces
1. This youtube channel
2. Angela yu course
3. react.dev
4. Open source projects to test ur skills
    a. gsoc orgs
    b. OS companies

Web3

1. Intro to blockchains
2. Cryptography
3. Solana architecture
4. Solana jargons (authority, owner)
5. PDAs
6. @solana/web3.js or gill
7. Solana wallet adapter
8. Data model
9. Token program
10. Defi (amm, dlmm, clmm, perps)
11. Rust easy
12. Rust advance
13. Anchor
14. common contracts (staking/escrow)
15. Indexing
16. MPC and Shamirs
17. Ad hoc web2 + web3
18. Partially centralized contracts

Projects

1. DEX
2. CEX
3. Wallet
4. Prediction market
5. Frontends/clients/tests
   for contracts (staking/escrow)

Resources

1. https://github.com/solana-foundation/curriculum
2. https://bitcoin.org/bitcoin.pdf
3. Rust -- Jon Gjengset

Devops

1. Bash/Terminal
2. VMs/Baremetal machines
3. Process management + Reverse proxies
4. Certificates and cert management
5. ASGs/MIGs
6. Containers and container runtimes
7. Docker
8. Kubernetes 1
9. Kubernetes 2
10. CI/CD
11. Monitoring/Observability
12. iac
13. CDNs + Object stores
14. Sandboxing/Firecracker

Projects

1. e2b
2. Replit
3. cloudflare workers

1. https://e2b.dev/blog
2. https://modal.com/blog